

Evolving MNK-landscapes with structural constraints

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Abstract—In this paper we propose a method for the generation of instances of the MNK-landscapes that maximize different measures used to characterize multi-objective problems. In contrast to previous approaches, the introduced algorithm works by modifying the neighborhood structure of the variables of the MNK-landscape while keeping fixed the local parameters of its functions. A variant of the algorithm is presented to deal with situations in which the exhaustive enumeration of search space is unfeasible. We show how the introduced method can be used to generate instances with an increased number of solutions in the Pareto front. Furthermore, we investigate whether direct optimization of the correlation between objectives can be used as an indirect method to increase the size of the Pareto fronts of the generated instances.

I. INTRODUCTION

Analyzing the relationships between the objectives to gain understanding about the characteristics and difficulty of multiobjective problems (MOPs) is a relevant research topic in multiobjective optimization (MOO) [4]. Research in this topic has been particularly active in the field of evolutionary algorithms [9], [13], [18], [19], [20]. Knowledge about the sign and strength of the correlations between the objectives can help to determine whether relationships between objectives are harmonious, conflicting or neutral. The analysis of correlations can be also used to propose objectives partitioning approaches [13] and more efficient optimizers [6], [7].

Many researchers have investigated the influence that correlations between objectives have in the complexity of the optimization problem and in the shape of the Pareto set [3], [10], [12], [25]. Problem test suites and problem generators have been also introduced where instances are created with desired degrees of correlations between the variables [9], [25], [26].

Some recent research has proposed techniques that exploit the dependencies between objectives and variables at the time of exploring the space of optimal solutions [6], [7]. In this framework, the detection of dependencies covers the case of (linear) correlations between objectives but also other type of (non-linear) dependencies. Furthermore, it is possible to capture, by means of the analysis of the dependencies, how particular subsets of variables influence specific objectives.

Therefore, a very complete picture of the type and range of interactions between decision variables and objectives can be learned and used by the decision maker.

There are two questions that are relevant for the design of enhanced algorithms that make the most of the dependencies between objectives and variables. The first, is to evaluate the effect that particular interactions between variables have in the parameters that describe a MOP, like the size of the optimal Pareto front, the number of Pareto fronts, etc [3], [25]. The second question is how to design instances that could maximize a given criterion of difficulty for a MOP while satisfying a given pattern of interactions between the decision variables. We call to this type of patterns *structural constraints*.

In this paper we take a first step toward the creation of instances that could serve to evaluate the techniques that model the interactions between objectives and instances. We propose the creation of MNK-landscape instances [2], [3] by directly optimizing some characteristics that describe the difficulty of the multi-objective problem. The MNK-landscape is a multi-objective variant of the NK-landscape [8] that has been previously applied to investigate different MOEAs [3], [5], [16].

The NK-landscape is a parametrized model of a fitness landscape that allows to explore the way in which the neighborhood structure and the strength of interactions between neighboring variables determine the ruggedness of the landscape. It defines an additively decomposable function of n components, where n is the number of variables. Each component of the landscape contributes to the total fitness of the function, and this contribution comes from local functions defined on each variable and an associated neighborhood of variables. The MNK-landscape considers multiple (M) NK-landscape models, each model is associated to an objective function of a MOP.

We introduce a novel approach in which the neighborhood structure of the MNK-landscape instance is modified while the parameters of the local functions are kept constant. The iterative generation of the instances is guided by an arbitrary measure of quality and executed using an optimization tech-

nique. Therefore, our goal is not to solve the MOP defined by the instances, but to create “optimal” instances in the sense of optimizing a given descriptor of the associated MOP. In doing so, we aim at investigating the role that the relationships between the variables have in different aspects related to the complexity of the MOP, like the size of the Pareto front. In contrast to other approaches that pre-define a measure to tune the landscape (e.g. the correlation between objectives), our method is general in the sense that it can be applied with different measures. The key idea is defining a search in the space of possible instances, and evaluating the quality of the instances based on criteria related to the complexity of the associated MOP.

The paper is organized as follows: In the next section, the MNK-landscape model is introduced. Section III discusses research related to our proposal. Our approach to evolve MNK-landscape instances of high complexity is explained in Section IV. Section V describes the experimental framework to evaluate our proposal and presents the numerical results. The main contributions of the paper are summarized in Section VI where some lines for future research are also discussed.

II. MNK-LANDSCAPE MODEL

A. NK fitness landscape

Let $\mathbf{X} = (X_1, \dots, X_n)$ denote a vector of discrete variables. We will use $\mathbf{x} = (x_1, \dots, x_n)$ to denote an assignment to the variables. S will denote a set of indices in $\{1, \dots, n\}$, and X_S (respectively x_S) a subset of the variables of $\mathbf{X}$ (respectively $\mathbf{x}$) determined by the indices in S .

The *NK fitness landscape model* is a parametrized model of a fitness landscape that allows to explore the way in which the neighborhood structure and the strength of the interactions between neighboring variables determine the ruggedness of the landscape. For given parameters, the problem consists of finding the global maximum of the function.

An NK fitness landscape [8] is defined by the following components:

- Number of variables, N .
- Number of neighbors per variable, K .
- A set of K neighbors $\Pi(X_j) \in \mathbf{X}$, for $X_j, j \in \{1, \dots, n\}$.
- A subfunction f_j defining a real value for each combination of values of X_j and $\Pi(X_j)$, $j \in \{1, \dots, n\}$.

The objective function f_{NK} to maximize is defined as:

$$f_{NK}(\mathbf{x}) = \sum_{j=1}^n f_j(x_j, \Pi(x_j)). \quad (1)$$

The complexity of the NK fitness landscape problem depends on all its components. For $K > 1$ it is NP-complete in the general case. The model has been extensively analyzed to study EAs and other heuristic algorithms [1], [15], [17], [24], [26].

B. MNK-landscape

The MNK-landscape model [2], [3] is an extension of the NK fitness landscape to the multiobjective case. An MNK-landscape is defined as a vector function mapping binary strings into M real numbers $\mathbf{f}(\cdot) = (f_1(\cdot), f_2(\cdot), \dots, f_M(\cdot)) : \mathcal{B}^n \leftarrow \mathcal{R}^M$, where n is the number of variables, M is the number of objectives, $f_i(\cdot)$ is the i -th objective function, and $\mathcal{B} = \{0, 1\}$. $\mathbf{K} = \{K_1, \dots, K_M\}$ is a set of integers where K_i is the size of the neighborhood in the i -th landscape. Each $f_i(\mathbf{x})$ is defined similarly to Equation (1) as:

$$f_i(\mathbf{x}) = \frac{1}{n} \sum_{j=1}^n f_{i,j}(x_j, \Pi_{i,j}(x_i)), \quad (2)$$

where the fitness contribution $f_{i,j}$ of variable x_j is a real number in $[0, 1]$ drawn from a uniform distribution.

III. RELATED WORK

Previous work has investigated the creation of instances that exhibit a-priori defined characteristics, like a given correlation between objectives [25]. Extensive research has been also done in the MNK-landscape. We briefly review some of this work in this section.

Knowles and Corne [9] introduced an instance generator for the multiobjective quadratic assignment problem (mQAP) and investigated the effect of correlations between the objectives. To enforce the correlations between the mQAP objectives, a correlation parameter c is used. Based on this fixed parameter, the instances are generated.

Aguirre et al. [3] present an exhaustive investigation of how the parameters of the MNK-landscape influence several characteristics of the fitness landscape, including the size of the Pareto front and the number of fronts. They extracted a number of valuable conclusions related to effects that the parameters of the MNK-landscape have in the complexity of the problems. They also compared NSGA-II and SPEA2 on the generated landscapes. Although the MNK-landscape offers numerous possibilities to better understand MOPs, the precise tuning of the correlation between objectives by exclusively using the parameters M and K seems to be a very difficult task.

Verel et al. [26], [25] extended the NK-model to design multiobjective functions with correlations. In the ρ MNK-landscapes, the epistatic structure is the same for all the objectives and the local fitness functions are not independent. Instead, the parameters of these functions follow a multivariate uniform law defined by a correlation matrix C which is used to tune the correlations between the objectives. In [26], a complete enumeration of the small ρ MNK-landscape instances was used to investigate the effect of the correlations in the characteristics of the Pareto front. In a recent work, the ρ MNK-landscape model has been used to investigate the influence of the parameters describing the landscapes in the behavior of different MOEAs [5], [16], showing the general usefulness of this type of models.

The ρ MNK-landscape and the strategy to generate instances introduced in this paper follow two opposite approaches. While the former modifies the correlations by manipulating the local functions and fixing the neighborhood structure, we use a different procedure. In our model, functions are the same for all objectives, but the interactions between variables are determined by changes to the local neighborhoods of the variables.

In [21], [22], instances of the NK-landscape model were created using a greedy algorithm similar to the one presented in this paper. However, the fitness function used to evaluate the instances was the success rate of different optimizers applied to solve the instances.

IV. TUNING THE MNK-LANDSCAPE

For our analysis, we will restrict the class of MNK-landscapes we deal with. The following constraints are enforced on the model:

- The number of neighbors per variable is the same for all functions f_i , i.e. $K_i = K$ for all $i \in \{1, \dots, M\}$ (as in [3], [25])
- For all objectives, the function table defining the values of the local functions, is the same, i.e. $f_{i,j}(\cdot) = g_j(\cdot)$ for all $i \in \{1, \dots, M\}$ (different from [3], [25]).

Notice that the function tables depend on the variables and not on the objectives. Nevertheless, for a given variable X_j its neighborhoods will be different for two different objectives, i.e. $\Pi_{i,j} \neq \Pi_{k,j}$ (as in [3], [25]).

The restrictions guarantee that the difference between the objectives will only depend on the particular types of interactions between the variables. For each objective, the local contribution of variable X_j to the global value of function f_i is the same, the only difference between objectives is in the particular subsets of variables that contribute to the fitness. Setting this constraint also allows us to compare different instances of the problem that share the same function tables but have different *neighborhood structures*. In other words, we can focus on the effect that the different patterns of neighborhood relationships have in the difficulty of MNK-landscapes.

Equation 3 shows an example of an MNK-landscape where $n = 6$, $M = 2$, and $K = 3$. The example is shown for adjacent neighborhood model but in this paper we consider a general neighborhood structure. The parameters of the model are shown in Table I.

$$\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), f_2(\mathbf{x})) \quad (3)$$

where f_1 and f_2 are defined as:

$$\begin{aligned} f_1(\mathbf{x}) &= \frac{1}{n} (g_1(x_1, x_2, x_3) + g_2(x_2, x_3, x_4) + g_3(x_3, x_4, x_5) \\ &\quad + g_4(x_4, x_5, x_6) + g_5(x_5, x_6, x_1) + g_6(x_6, x_1, x_2)) \\ f_2(\mathbf{x}) &= \frac{1}{n} (g_1(x_1, x_4, x_6) + g_2(x_2, x_1, x_6) + g_3(x_3, x_6, x_2) \\ &\quad + g_4(x_4, x_1, x_3) + g_5(x_5, x_1, x_2) + g_6(x_6, x_5, x_4)) \end{aligned}$$

	Configurations							
	000	001	010	011	100	101	110	111
g_1	0.01	0.43	0.21	0.15	0.88	0.91	0.03	0.69
g_2	0.84	0.13	0.18	0.56	0.32	0.11	0.53	0.29
g_3	0.96	0.38	0.16	0.04	0.08	0.86	0.90	0.54
g_4	0.45	0.17	0.85	0.68	0.52	0.37	0.76	0.92
g_5	0.73	0.80	0.66	0.42	0.65	0.82	0.29	0.24
g_6	0.59	0.79	0.53	0.80	0.57	0.73	0.01	0.17

TABLE I
EXAMPLE OF ASSIGNMENT TO THE LOCAL FUNCTIONS.

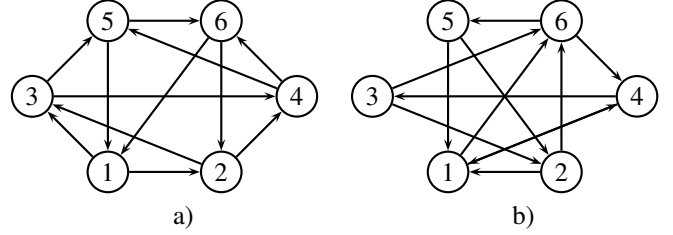


Fig. 1. Graphical representation of the neighborhood structures for functions f_1 and f_2 in the MNK-landscape defined by function (3) a) Neighborhood structure for f_1 ; b) Neighborhood structure for f_2 .

A. Graphical representation of MNK-landscapes

In the graphical representation of the MNK-landscape, we will assign a graph $G_i = (V, A_i)$ to each objective of the model where V is a set of vertices and A_i is a set of arcs. In the set of vertices, there is one vertex v_j associated to each variable X_j . In A_i , there exists one arc from vertex j to vertex k if and only if $X_k \in \Pi_{i,j}$.

As a consequence of the definition, the outdegree of all vertices is K , and the indegree can have a maximum value of $n - 1$. Figure 1 shows the graphs corresponding to the neighborhoods of the two objectives of the MNK-landscape defined in Equation 3.

The graphical representation is important because, once the subfunctions values have been fixed, we can try to explain the complexity of the instances in terms of the topological properties of the graphs they define. These properties are also important for algorithms based on the detection and use of interactions between variables [11], [14], [23].

B. Generation of the MNK-landscape instances

Firstly, we introduce some notation needed for the analysis presented in this section.

We assume that the underlying dominance structure in the MNK-landscape is given by the Pareto dominance relation “ $\mathbf{y}$ dominates $\mathbf{x}$ ” that is defined as $\forall \mathbf{x}, \mathbf{y} \in X, \mathbf{x} \preceq_{\mathcal{F}} \mathbf{y} \iff f_i(\mathbf{x}) \leq f_i(\mathbf{y}) \forall i$, and $\exists i$ such that $f_i(\mathbf{x}) < f_i(\mathbf{y})$ where $\mathcal{F} = \{f_1, \dots, f_M\}$.

The Pareto (optimal) set is given as $\{\mathbf{x} \in \mathcal{B}^n \mid \nexists \mathbf{y} \in \mathcal{B}^n \setminus \{\mathbf{x}\} \text{ such that } \mathbf{x} \preceq_{\mathcal{F}} \mathbf{y}\}$.

Our approach consists on increasing the complexity of an instance by re-arranging the neighborhood structure of the variables, without changing the local parameters of the model.

A hill climbing procedure is used to change, in a parsimonious way, the structure of the variables neighborhood of MNK-landscape in such a way that a measure that describes the complexity of the MOP is maximized.

Modifying an instance to maximize its difficulty allows us to study the process by which instances become more complex. This can not be done by directly generating difficult instances. Since the difficulty of an instance will depend to some extent on the solver used, we investigate different features that characterize the MOP.

Algorithm 1: Greedy MNK-landscape optimizer

```

1  Generate a random MNK-landscape instance.
2  Set  $MaxVal$  to the fitness value computed for the
   instance.
3  do {
4    For each objective  $i$ , each node  $j$ , and each neighbor
       $k$  in  $\Pi_{i,j}$ 
5      For all possible  $l \in \{1, \dots, n\} \setminus \Pi_{i,j}$ 
6        Replace  $k$  by  $l$  in  $\Pi_{i,j}$ 
7        Compute the fitness value  $CurrVal$  from the
          new MNK-landscape
8        If  $CurrVal > MaxVal$ 
9          Then  $MaxVal = CurrVal$ 
10         Else Replace  $l$  by  $k$  in  $\Pi_{i,j}$ 
11 } until Termination criteria are met

```

Algorithm 1 shows the pseudocode of the algorithm used for optimizing (maximization) the characteristics of the MNK-landscape. In Step 1, first the neighborhood structure of each vertex is randomly selected using a uniform distribution without replacement on the set of vertices. Values for the function tables are randomly generated from a uniform distribution.

The termination criteria of the algorithm can be a maximum number of iterations or no improvement in the fitness function (the chosen characteristic of the complexity of the MOO generated) used to guide the search. One key issue is how the computation of the fitness function is done. We consider two different cases: 1) The exact case: When it is possible to compute an exact value of the measure, and 2) The approximate case: when, due to the complexity associated to the computation of the fitness function, an approximation has to be computed. Examples of measures that we use in our experiments are the size of the Pareto front, and the correlation between objectives.

The evaluation of the complete search space required by Algorithm 1 is only feasible for small values of n . However, we can also optimize instances for more realistic values of n . This can be done in two different ways: 1) By using a sample of solutions from the search space as the basis for computing the fitness values used to guide the search for optimal instances. 2) By randomly selecting a neighborhood, as well as randomly selecting the changes to the variables, as a way to avoid the exhaustive inspection of the neighborhoods described in steps 4 and 5 of Algorithm 1.

Depending on the parameters used (e.g. size of sample of

solutions, number of trials) the stochastic search variant can be much faster than Algorithm 1. However, for certain measures of difficulty (e.g. those related to the computation of the Pareto front) the approximation provided by the sample of solutions can be misleading. For instance, Pareto front approximations computed from a small sample of the search space will not likely provide accurate descriptors of the instance complexity.

V. EXPERIMENTS

The description of the experiments is divided into two parts. In the first part, we explain the results of the exact method for the situation in which the total search space is enumerated. In the second part, we analyze the results for the approximate variants of the algorithm.

A. Results of the exact method

The goals of the experiments included in this part are twofold. The first goal is to evaluate whether the introduced method is able to generate instances that maximize different parameters associated to the complexity of the MOP, and to determine what the influence of the MNK-landscape parameters in the characteristics of the instances generated is. We select the size of the Pareto front and the correlation between the objectives as the characteristics to investigate. The choice of these two characteristics is due to their important influence in the behavior of MOEAs and also to the link between them. The fact that the shape of the PF can be greatly influenced by the correlation between the objectives has been previously reported [10], [19]. Therefore, a second goal of our experiments is to investigate whether the minimization of the correlation between objectives can be used as an indirect way to maximize the number of solutions in the Pareto front. We compare instances generated by direct maximization of the size of the Pareto front with instances generated using this indirect approach.

We use $n = 10, 12$, $M = 2$, $k = 1, \dots, 4$. The number of steps for Algorithm 1 was $ns = 25$. For each combination of parameters, l executions of the algorithm were run ($l = 30$ for $n = 10$, and $l = 10$ for $n = 12$), using in each case a different initialization for the neighborhood structure of the second objective. We kept one of the objectives fixed, i.e. no changes were applied to its neighborhood structure.

Figure 2 shows the distribution of the objectives values and the Pareto front of a randomly generated instance of the MNK-landscape. Figure 3 shows similar data for the instance obtained as a result of applying the algorithm for optimizing the size of the Pareto front ($ns = 25$). The fitness values of f_1 are identical for the two instances. However, the local changes applied by the algorithm to the neighborhood structure of objective f_2 determine that the number of solutions in the Pareto front for the generated instance is higher. The key point here is that the changes in the objective f_2 that lead to a more populated Pareto front in the generated instance are all local and in every step they involve only two variables. We can see these changes as small perturbations or rearrangements in the neighborhood structure of the objective.

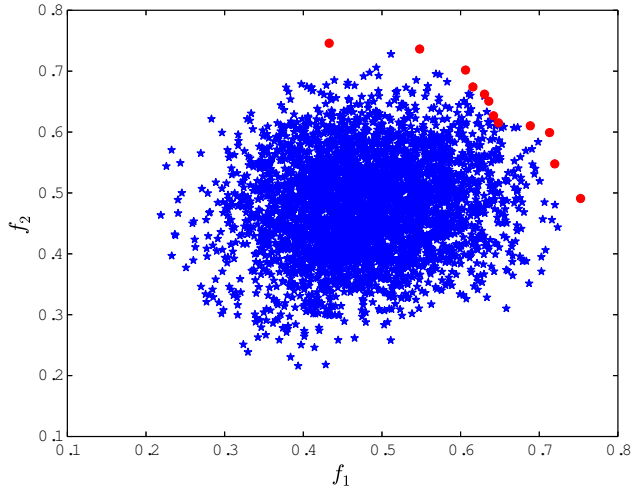


Fig. 2. Distribution of the objectives values of an initial MNK-landscape instance for $n = 12$, $M = 2$, $k = 1$. The points in the Pareto front are represented using circles and marked in red.

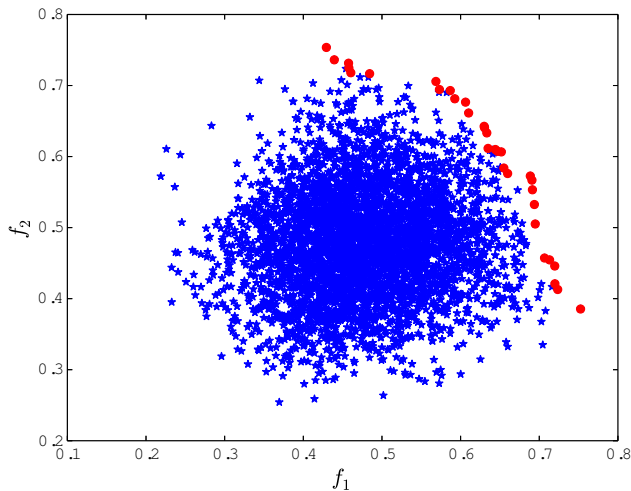


Fig. 3. Distribution of the objectives values of the instance generated by Algorithm 1 from the MNK-landscape shown in Figure 2. The goal of the optimization has been to maximize the size of the Pareto front.

We evaluate the influence that the parameters of the MNK-landscape (n and k) have in the characteristics of the instances constructed and in the behavior of the algorithm. Figures 4, 5, and 6 respectively show the average value of the measure optimized for the problems of: 1) Minimizing the correlation between the objectives. 2) Maximizing the correlation between the objectives. 3) Maximizing the number of points in the Pareto front. In the figures it can be seen that the influence the parameters have depends on the measures optimized. For the problem of minimizing the correlation, there is a clear difference between $n = 10, k = 1$ and the rest of MNK-landscapes. The initial correlation is very high when each

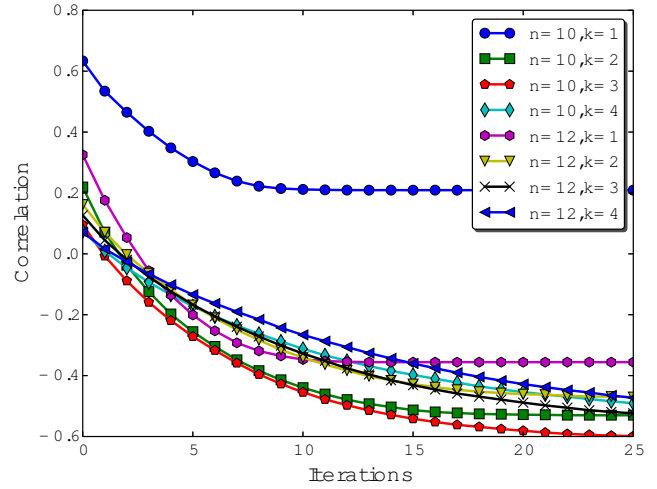


Fig. 4. Influence of the parameters n and k of the MNK-landscape for the problem of minimizing the correlation between objectives. Average correlation at each step of the algorithm.

variable depends on a single neighbor, and this correlation is very difficult to reduce.

The problem of maximizing the correlation is much easier, particularly for $k = 1$. But as k grows the algorithm needs more steps to increase the correlations. Finally, for the problem of directly maximizing the size of the Pareto front it is not clear the influence of the parameter k . The least and the most populated Pareto fronts are obtained for $k = 1$, although for different values of n . The combination of the two parameters seems to play a key role for this measure.

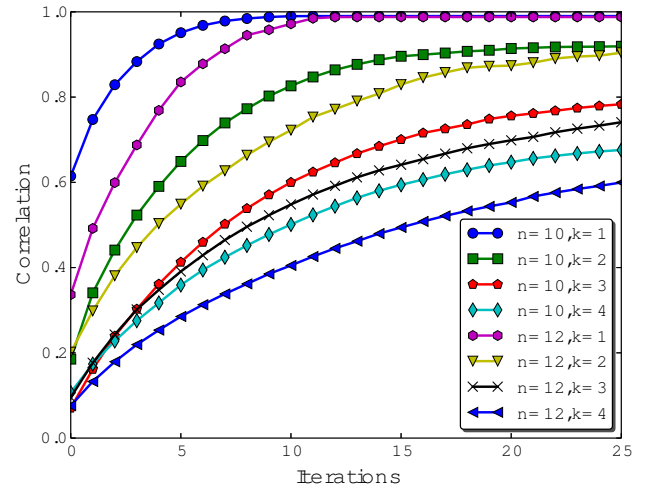


Fig. 5. Influence of the parameters n and k of the MNK-landscape for the problem of maximizing the correlation between objectives. Average correlation at each step of the algorithm.

A question that has been extensively analyzed in MOPs is the relationship between the correlation of the objectives and

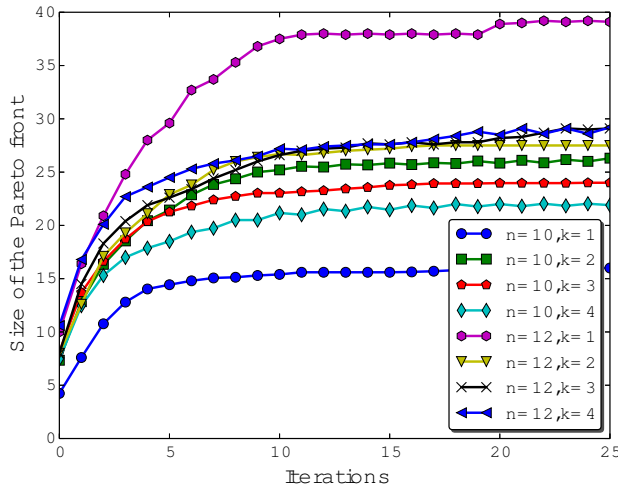


Fig. 6. Influence of the parameters n and k of the MNK-landscape for the problem of maximizing the number of points in the Pareto front. Average size of the Pareto front at each step of the algorithm.

the shape of the Pareto front, particularly when the objectives exhibit a conflicting relationship. We wanted to determine if instances constructed according to the minimization of the correlation had, on average, a more populated Pareto front than random instances. To do this we compare the additive difference in the size of the Pareto sets of the instances obtained using the three measures explained before. The results of this analysis, for different values of n and k , are summarized in Table II.

It can be seen in Table II that the minimization of the correlation can lead to an increase in the size of the Pareto front of the generated instances. The increase is only a fraction of the increment obtained by maximizing the size of the Pareto front directly. However, in situations where the exact computation of the Pareto front is not possible, minimizing the correlation arises as a suitable alternative to generate instances with populated Pareto fronts. As expected, it can be also observed in Table II that maximizing the correlation reduces the number of points in the Pareto front. Actually, in instances for which the correlation has been maximized the size of the Pareto front is 1.

n	k	$Min.Corr$	$Max.Corr$	$Max.PF$
10	1	4.0000	-4.0333	11.7667
	2	9.0333	-6.7667	18.9667
	3	6.6000	-4.7333	15.8667
	4	4.0667	-2.7333	14.4333
12	1	8.5000	-7.600	29.1000
	2	7.1000	-3.9000	19.5000
	3	4.9000	-3.0000	21.1000
	4	6.6000	-3.5000	18.6000

TABLE II

INCREASE IN THE SIZE OF THE PARETO FRONTS DUE TO THE APPLICATION OF THE ALGORITHM ON A RANDOM INSTANCE OF THE MNK-LANDSCAPE AND USING DIFFERENT MEASURES TO GUIDE THE SEARCH.

B. Results of the approximate method

In this section we present results of two variants of the approximate algorithm in different scenarios. First, we apply a variant of Algorithm 1 that stochastically searches in the neighborhoods but using the complete enumeration of the search space. With this algorithm we address the problem of maximization of the size of the Pareto front and the problem of minimizing the number of fronts of the MNK-landscape. Additionally, we apply the algorithm to problems with $M = (2, 3)$ objectives and allow the method to optimize any of the objectives in every step of the algorithm. Then we use another variant of the algorithm that in addition to stochastically search in the neighborhoods, uses a sample of solutions to avoid the complete enumeration of the search space. The algorithm is applied to the minimization of the correlation, and in this case only one of the two objectives is optimized.

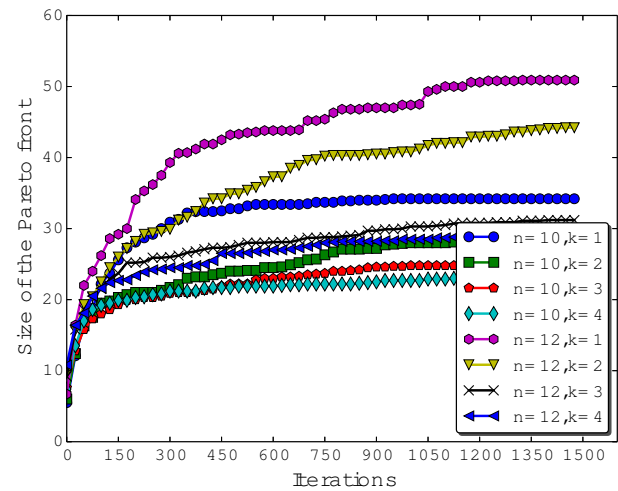


Fig. 7. Maximization of the Pareto front size. Two objectives.

Figures 7 and 8 show the results of the algorithm in the problem of maximizing the size of the Pareto set for different values of n and k . Figure 7 shows the case of two objectives and in Figure 8 results are presented for a problem with three objectives. The number of steps allowed to the algorithm was $ns = 1500$.

It can be seen in figures 7 and 8 that the algorithm is able to achieve a dramatic improvement in the number of solutions in the Pareto front. This is particularly evident for $n = 12, k = (1, 2)$. As the number of objectives grows from two to three, and as expected, it is possible to obtain instances with more populated Pareto fronts. As k increases it becomes more difficult to find instances with a higher number of solutions in the Pareto front as previously reported in [3].

The other problem we address using this variant of the algorithm, and with the same parameters, is the minimization of the number of fronts. This is a relevant question because the number of fronts is another property that can serve to

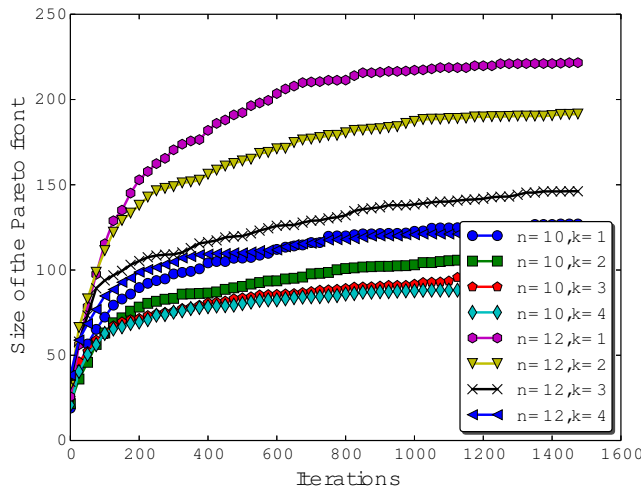


Fig. 8. Maximization of the Pareto front size. Three objectives.

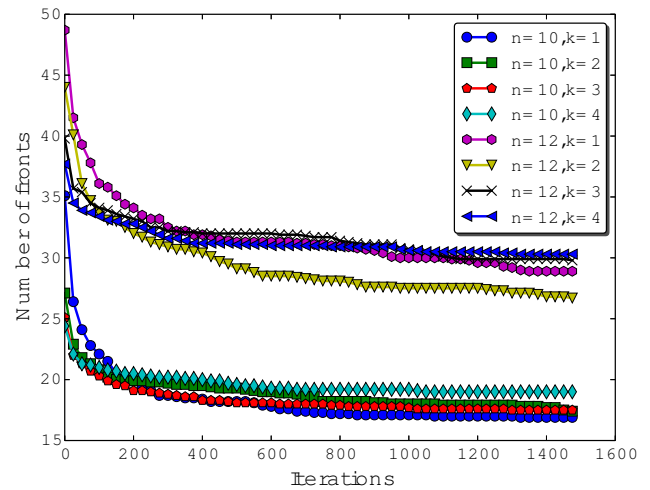


Fig. 10. Minimization of the number of fronts. Three objectives.

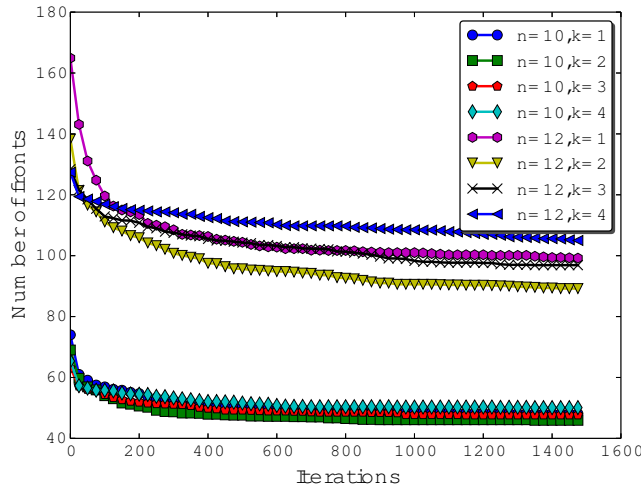


Fig. 9. Minimization of the number of fronts. Two objectives.

characterize MOPs and can be used to evaluate the behavior of MOEAs. For the MNK-landscape, it was also previously investigated in [3].

Figures 9 and 10 respectively show the results of the algorithm for this problem, when applied to the generation of instances with two and three objectives. It can be observed in the figures that the algorithm is able to find at each step better instances. However, as the number of fronts is reduced, it takes more steps of the algorithm to find better instances. Also in this case, a higher level of epistasis in the landscape implies that it is more difficult to reduce the number of fronts.

In the next experiment, the stochastic variant of Algorithm 1 is applied to the minimization of the correlation for an MNK-landscape with parameters $n = 50$, $M = 2$, $k = 1, \dots, 4$. Only one of the objectives is modified in each step, the

algorithm uses a sample size of 1024 solutions, and the number of trials (potential changes evaluated in each step of the algorithm to select the next arc replacement) is 50.

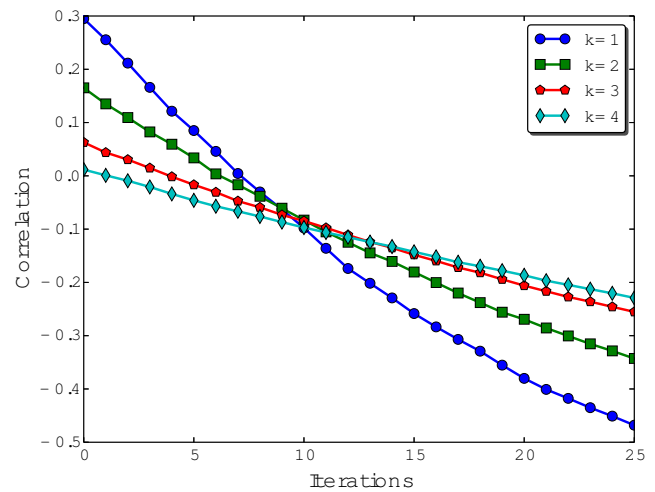


Fig. 11. Influence of k in the problem of minimizing the correlation between objectives. Average correlation at each step of the algorithm.

Figure 11 shows the average correlation at each step of the algorithm for different values of k . It can be seen in the figure that the algorithm is able to reduce the correlation for all values of k . However, in contrast with the results shown in Figure 11, the best results are achieved for $k = 1$. As k is increased, it is more difficult for the algorithm to reduce the correlation. One reason that could explain this behavior is that, for a fixed number of 50 trials, the relatively small neighborhood structure of the MNK-landscape with $k = 1$ can be inspected very well. However, as k grows, only a fraction of the arcs in the neighborhood structure of the model can

be inspected. Therefore, many moves that could improve the optimized measure are overseen. One solution for this problem would be to set the number of trials as a function of k .

VI. CONCLUSIONS

In this paper we have introduced a new optimization-based algorithm for generating MNK-landscapes instances that optimize a number of characteristics related to the difficulty of the MOP. We have shown that by focusing on changes in the interactions between the variables while keeping the parameters of the functions fixed, it is possible to produce instances of the MNK-landscape tuned according to some criterion specified by the user.

In our experiments we observed that it is relatively easy to obtain an increase in the correlations between the variables by changing the structure of the neighborhoods. However, achieving clearly conflicting objectives by means of reducing the correlations is a harder task.

The introduced algorithm can be applied to “tune” instances by forcing them to satisfy dissimilar constraints captured in the fitness function used to guide the search in the space of the instances. For example, we could search for instances that minimize as much as possible the correlation while keeping a small Pareto front. The question of constructing such instances is transformed in an optimization problem for which the challenge is to define an appropriate objective function that combines the different target characteristics. The key advantage over methods to generate instances that are focused on a particular measure (e.g. the correlation) is that the criterion to evaluate the suitability of the instances is open. Thus, our method can provide another perspective to study the relationship between the different parameters that characterize MOPs.

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